Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which steps are the same for EKF and SPKF for parameter estimation? (Select all that apply.)

1 / 1 point

- [x] Step 1a: Parameter prediction time update.

Correct
Yes. The two filters implement this step the same way.

- [x] Step 1b: Prediction error covariance time update.

Correct
Yes. The two filters implement this step the same way.

- [ ] Step 1c: Output prediction.
- [ ] Step 2a: Estimator gain matrix.
- [x] Step 2b: Parameter estimate measurement update.

Correct
Yes. The two filters implement this step the same way.

- [x] Step 2c: Estimate error covariance measurement update.

Correct
Yes. The two filters implement this step the same way.

2. Which of the following derivative matrices are required when implementing EKF for parameter estimation? (Select all that apply.)

1 / 1 point

- [ ] The $\hat{A}_k^\theta$ matrix.
- [ ] The $\hat{B}_k^\theta$ matrix.
- [x] The $\hat{C}_k^\theta$ matrix.

Correct
Yes. This matrix is required.

- [x] The $\hat{D}_k^\theta$ matrix.

Correct
Yes. This matrix is required.

3. Which of the following derivatives must be computed to implement the steps of the EKF for parameter estimation? (Select all that apply.)

1 / 1 point

- [ ] The derivative df/dx.
- [x] The derivative $df/d\theta$.

Correct
Yes. This is part of $\hat{C}_k^\theta$.

- [ ] The derivative dh/dx.
- [x] The derivative $dh/d\theta$.

Correct
Yes. This is part of $\hat{C}_k^\theta$.